VILAS GANNARAM

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Summary

Fullstack Software Engineer with 3.5+ years shipping production systems across genomics/clinical data platforms and agentic AI products — from React/Next.js/Svelte frontends to Python/Node backends and the infra that deploys them.

Technical Skills

- **Programming Languages:** JavaScript(ES6+), TypeScript, Python, Go/GoLang, SQL, Bash
- **Frontend & UI:** React.js, Next.js, Redux, Zustand, SvelteKit, shadCN/UI, AntD, HTML5, CSS3, TailwindCSS, D3, Plotly
- **Backend & Data:** Node.js/Express.js, FastAPI, RESTful API, GraphQL, PostgreSQL, SQLite3, DuckDB, Redis
- **AI/LLM & Agentic AI:** Google ADK, LangChain, LangGraph, SDKs(GenAI, Vercel, OpenAI), Qdrant Vector DB
- **Tools & DevOps:** Claude Code, Git/GitHub, GitHub Actions, Azure, Linux, Nginx, Docker, Docker Compose

Work Experience

Aganitha AI
Hyderabad, IND

Fullstack Developer 1
Sept. 2025 - Present

Clinical Case Knowledge Portal | Internal R&D

- Built a clinical case knowledge portal (Astro/Svelte/TypeScript) over a schema-validated JSON data pipeline, powering case exploration and presentation for internal R&D review.
- Designed a provider-agnostic LLM chat layer (Node.js/Express/LangChain) — a tool-calling agent that answers case questions over deterministic data primitives, swapping between Anthropic/OpenAI/Google with zero code changes.

Igniva v2 — Aganitha's Agentic AI Platform | Product

- Evolved the Next.js/Typescript frontend for Igniva v2, scaling the multi-agent workflow platform (Google ADK/Python) to orchestrate scientific workflows, which involves tool calls (GROMACS, Boltz, AlphaFold...).
- Built end-to-end features(React UI to FastAPI) Share Chat and Continue Chat: sharing a conversation as a read-only snapshot, that recipients can fork into an independent session by deep-cloning full state (messages, agent events, artifacts).
- Built custom MarkdownRenderer (React.js/TypeScript) — a fenced-code-block rendering package featuring core, chem, bio, and data submodules —published to private npm registry — adopted across multiple Aganitha projects

Aganitha AI
Hyderabad, IND

Jr Fullstack Developer 2
Sept. 2024 - August 2025

Igniva v1 — Aganitha's Agentic AI Platform | Product

- Owned the Next.js/TypeScript frontend for Igniva's initial release (Google ADK/Python), building the real-time chat UI with SSE streaming for agent steps and tool calls, alongside a typed SDK client (OpenAPITool) and an MDX-based content registry.
- Built multi-agent orchestration on the Python backend (Google ADK) with schema-defined structured responses, developing specialist agents (IgnivaLab , MarketInsights, and LiteratureSearch)
- Developed a user-facing RAG-based agent catalog with semantic search (GenAI embeddings, Qdrant) so users find the right specialist agent from a natural-language query.

Research Data Products — GWAS Tools for Global Genomics Company | Client

- Genome Browser — Built full-stack app (SvelteKit/Node.js/TypeScript) with a DuckDB layer over large Parquet datasets (SNP, genotype, imputed, transcript, gene-location), exposed via REST APIs, visualized with D3.js, deployed with Kubernetes(minikube)

Aganitha AI
Hyderabad, IND

Jr Fullstack Developer 1
Sept. 2023 - August 2024

Research Data Products — GWAS Tools for Global Genomics Company | Client

- Burden Reports — Owned React.js/Apollo/TypeScript frontend with D3.js visualizations for gene-level burden test results; co-designed GraphQL schema (Hasura/PostgreSQL) over a ~300GB, multi-million-record dataset, deployed with Docker/Nginx
- Entity Search — Built a Python fastAPI/DuckDB service for gene and phenotype search over OpenTargets data, and published the UI component to a private npm registry for reuse across GWAS tools, & the fastAPI to Private Docker registry

Aganitha AI Hyderabad, IND

Fullstack Developer Intern March 2023 - August 2023

Company Website | Internal

- Shipped new pages and maintained the production 11ty/Nunjucks/TypeScript JAMstack site.
- Led the migration off JAMstack to CMS(WordPress/Elementor), unblocking non-technical teams to publish content independently — containerized the deployment and swapped Caddy for OpenLiteSpeed, landing an A+ GTmetrix score.

Projects

Lynkr | URL Shortener

- Designed & built a URL shortener in Go/Golang (Chi router) with PostgreSQL + Redis cache-aside, SSR UI via Go templates.
- Load-tested with k6 (100 concurrent users, 10k requests): 0% errors, ~260ms median / ~339ms p95, 362 req/s.
- Github: https://github.com/vilas-gannaram/Lynkr

Basemark | Custom Markdown Renderer

- Designed and built Basemark, a framework-agnostic markdown-directive engine (TypeScript/pnpm/Turbo monorepo) that resolves custom directives into interactive Web Components, built as an authoring surface for AI-generated content.
- Published 6 npm packages (@basemark/core, react, charts, bio, common, cli) spanning framework bindings, a chart pack, and a bioinformatics visualization pack (LocusZoom.js, ProtVista, Genome Browser).
- Github: https://github.com/vilas-gannaram/basemark, NPM: https://www.npmjs.com/org/basemark

Education & Courses

Osmania University — Hyderabad, India | 2019 - 2022

Bachelor's Degree in Arts | 8.3/10 CGPA

The Web Developer Bootcamp — Colt Steele | UDEMY | 2022

Completed comprehensive training in modern JavaScript (ES6+), asynchronous programming, and full-stack development to build a professional technical foundation.

Certifications — Q4 2021

- Azure Fundamentals (AZ-900)
- Azure AI Fundamentals (AI-900)
- Azure Data Fundamentals (DP-900)